Joint Placement-Policy Optimisation for Charging Stations in RMFS

A Constrained RSM and Taguchi Robustness Approach

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Problem Statement — The Single-Axis Trap

What an RMFS warehouse needs to decide:

- **Where** to install charging stations (a CapEx commitment).
- **When** a robot should go charge, and how full to charge (an OpEx parameter).

The literature treats them separately:

- Placement studies fix the policy and ask "where?"
- Policy studies fix the placement and ask "when?"

The hypothesis:

- Placement and policy *interact*.
- Joint optimisation reveals failure modes invisible to either axis alone — e.g., **deadlock manifolds** that emerge from the combination of policy thresholds and charger geometry.

Our research question: Given a joint design space (placement $\times$ policy $\times$ budget), what is the operationally robust optimum, and what statistical machinery surfaces it?

Research Contributions

Methodological:

- Treatment of discovered deadlock zones as *infeasibility constraints* rather than missing data — preserves statistical defensibility, turns failures into research findings.
- Single-seed deterministic DoE protocol with formal-test-free residual diagnostics for simulation environments.

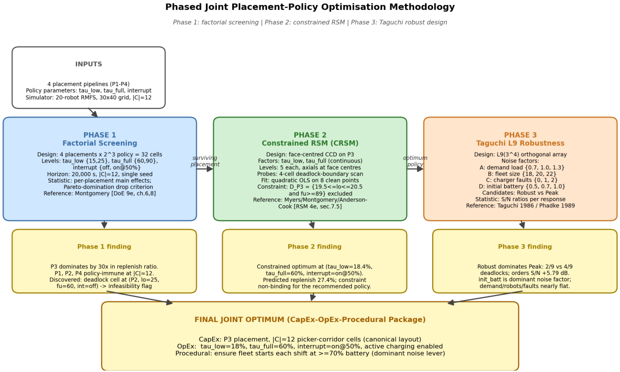
Empirical:

- Placement quality dominates charger-budget by $\sim 30\times$ on long-horizon replenish ratio.
- Constrained-RSM optimum is also the Taguchi-robust optimum (rare in practice; both axes pointing the same direction).
- Initial battery state at shift start is the *dominant* noise lever, not demand swing or fleet attrition.

Managerial (decision-grade output):

- A CapEx-OpEx-Procedural triple (placement, policy, operating practice) directly deployable in industry, each tier traceable to a specific experimental phase and defended by its own statistical artefact.

Methodology — Phased DoE Framework



Phase 1 — Screening.
Four placements (P_1 Set-Cover, P_2 Affinity-Propagation, P_3 Picker-Station, P_4 Perimeter) at default policy. Long-horizon ($T = 100\,000\,\mathrm{s}$) + budget sweep.

Phase 2 — Constrained RSM.
Face-centred CCD over policy thresholds on surviving placement. Discovered deadlock manifold treated as infeasibility constraint.

Phase 2b — Cross-placement RSM.
21-cell design across all four placements + Michaelis–Menten saturation fit on $|C|$.

Phase 3 — Taguchi L_9.
Four operational noise factors $\times$ S/N analysis comparing Phase-1 best vs Phase-2 RSM optimum.

Theory — Design of Experiments and Response Surfaces

Design of Experiments (DoE). A statistical framework (Fisher, 1925; refined by Box & Wilson, 1951) for systematically varying input factors and analysing their effect on a response variable, using as few experiments as possible while preserving statistical power.

Response Surface Methodology (RSM). A second-stage DoE technique that fits a smooth polynomial model around an operating region:

$$y = \beta_0 + \sum_i \beta_i x_i + \sum_i \beta_{ii} x_i^2 + \sum_{i<j} \beta_{ij} x_i x_j + \varepsilon$$

The fitted surface enables analytical optimisation (gradient $= 0$, Hessian negative-definite) within the design region. Standard textbook reference: Myers, Montgomery & Anderson-Cook (2016).

Central Composite Design (CCD). The standard RSM design — combines:

- 2^k factorial corners (estimate main effects + interactions)
- $2k$ axial points (estimate curvature)
- n_c centre replicates (estimate pure error)

Face-centred CCD ($\alpha = 1$): axial points on the cube faces rather than outside. Chosen here because our factor bounds are operational limits, not just numerical; pushing axial points beyond them is physically meaningless.

Theory — Why CCD, and Not Box-Behnken or Plackett-Burman?

Design	Strength	Why not chosen here
Plackett-Burman	Maximum screening power per run	Only main effects; no curvature, no interaction — insufficient for an optimisation question.
2^k full factorial	Captures main effects + all interactions	Linear model only; cannot locate a smooth interior optimum.
Box-Behnken	Avoids extreme-corner points	Doesn't sample the face/corner combinations where our deadlock manifolds live. Would have missed the constraint surface entirely.
Face-centred CCD	Quadratic surface + corner coverage + boundary respect	**Chosen.** Reuses the Phase-1 factorial corners, samples the boundary where deadlock occurs, and supports analytical optimisation.

Bottom line: face-centred CCD is the only standard design that simultaneously supports quadratic curvature fitting, respects physical factor bounds, and reuses the screening corners we already ran in Phase 1.

Theory — Constrained RSM and the Deadlock Manifold

Standard RSM assumes the response surface is smooth and differentiable across the design region. But in our simulator, certain $(\tau_{\text{low}}, \tau_{\text{full}})$ combinations trigger **deadlock** — a discrete state in which robots block each other indefinitely. The response collapses to near-zero at these points.

Three options for handling discovered deadlock points:

1. Drop them as outliers
   $\Rightarrow$ statistically illegitimate; biases the fitted surface.
2. Treat them as low-response observations
   $\Rightarrow$ the quadratic fit becomes a worse model of the smooth region.
3. **Treat them as an infeasibility region.**
   $\Rightarrow$ fit the smooth model on the feasible subset; optimise subject to constraint $g(\mathbf{x}) \geq 0$, where g separates safe from deadlock zones.

Constrained-RSM formulation (Myers, Montgomery & Anderson-Cook, 2016):

$$\max_{\mathbf{x} \in \mathcal{D}} \hat{y}(\mathbf{x}) \quad \text{s.t.} \quad g(\mathbf{x}) \geq 0$$

Methodological contribution: the constraint surface $g(\mathbf{x}) = 0$ *is itself a research finding* — it characterises the deadlock manifold rather than hiding it.

Theory — Taguchi Robust Parameter Design

Core idea (Taguchi, 1986): A design is *robust* if its performance is insensitive to uncontrollable noise factors. Optimising for robustness $\neq$ optimising for mean performance — and the two can disagree.

Orthogonal arrays (e.g. L_9, L_{18}, L_{27}) sample noise-factor combinations in a balanced way, allowing the noise envelope to be characterised in few runs. The $L_9(3^4)$ design probes four 3-level factors in only 9 runs (full factorial would need $3^4 = 81$).

Signal-to-Noise (S/N) ratios convert mean and variance into a single optimisation target:

$$\text{Larger-the-better (LTB):} \quad \eta_{\text{LTB}} = -10\log_{10}\left(\tfrac{1}{n}\textstyle\sum_i 1/y_i^2\right)$$

$$\text{Smaller-the-better (STB):} \quad \eta_{\text{STB}} = -10\log_{10}\left(\tfrac{1}{n}\textstyle\sum_i y_i^2\right)$$

In our Phase 3: y = orders completed (LTB), y = energy per order (STB). Higher S/N is always better.

Our four noise factors: demand load, fleet size, charger faults, initial battery state.

The Four Placement Algorithms

Pipeline	Algorithm	Logic
P_1	**Set Cover Approximation** (greedy)	Classic NP-hard set-cover problem; covers high-traffic cells within radius ρ. Spreads chargers across the warehouse.
P_2	**Affinity Propagation** (Frey & Dueck, 2007)	Message-passing clustering of high-traffic cells into exemplars. Self-determining number of clusters.
P_3	**Picker-Station Opportunistic**	Placement aligned to picker-station corridors. Robots charge during natural dwell time at stations $\Rightarrow$ no active charge trips.
P_4	**Perimeter Isolation**	Charger row on warehouse perimeter near replenishment stations. Acts as a control.

Two design philosophies: P_1, P_2, P_4 *decouple* charging from the work flow (robots must leave their task to charge). P_3 *couples* charging to the work flow (charging happens passively while the robot is already at a station). The performance gap between these philosophies is what Phase 1 quantifies.

Energy Model — The Two-Channel Architecture

Battery drain is split into two physically independent channels:

Electronics channel (mass-independent, constant 90 J/s):

$$E_{\text{idle}}^{(\Delta t)} = P_{\text{drain}} \cdot \Delta t$$

CPU, sensors, radio, lights. Active whenever not charging, regardless of payload.

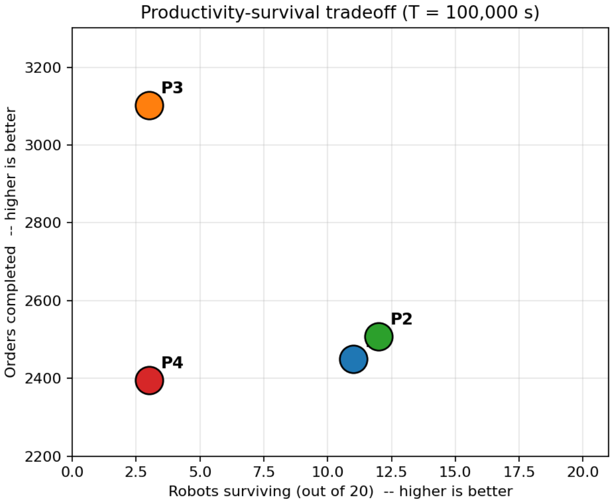
Mechanical channel (mass-scaled, motion-only):

$$\begin{aligned}
E_{\text{const}} &= (m + m_L)\, g\mu\, v\, \Delta t \qquad \text{(rolling friction)} \\
E_{\text{accel}} &= (m + m_L)\, (a\, k_{\text{ir}} + g\mu)\, v\, \Delta t \\
E_{\text{rot}} &= \tfrac{1}{6}(m + m_L)(L^2 + W^2)\, \frac{\theta^2}{\tau_{\text{rot}}^2} + (m + m_L) g\mu r\theta \\
E_{\text{lift}} &= m_L \cdot g \cdot h_{\text{lift}}
\end{aligned}$$

Why the split matters: a heavier robot standing still costs the same as an empty one (electronics-only). The cost of carrying a pod is paid only while the wheels are turning.

Battery spec (commercial AGV): $E_{\text{cap}} = 1.8\,\text{kWh} = 6.48\,\text{MJ}$; $P_{\text{charge}} = 397.6\,\text{W}$ ($= 28\,\text{A} \times 14.2\,\text{V}$); $5\,\%/\text{hr}$ idle drain $\Rightarrow$ $90\,\text{J/s}$.

Phase 1 — Placement Dominates Budget by an Order of Magnitude

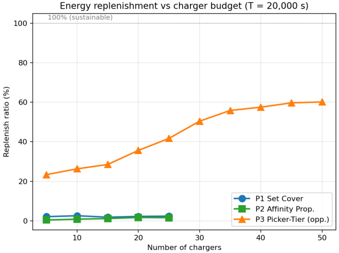


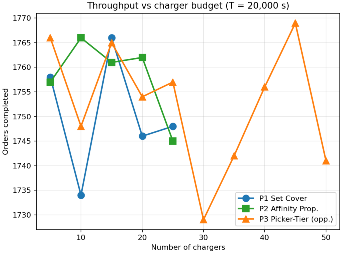
Long-horizon result ($T = 100\,000$ s, $|C| = 12$, $N_r = 20$).

- P_3 (Picker-Tier) completes the most orders but loses fleet integrity.
- P_2 (Affinity Propagation) preserves fleet but completes fewer orders.
- P_1 sits between them.
- P_4 (Perimeter) is Pareto-dominated.
- **No placement at $|C| = 12$ is operationally sustainable under default policy.**

Replenish ratio gap: P_3 achieves $\sim 30\times$ the replenishment-vs-consumption ratio of $P_1/P_2/P_4$.

Phase 1 — Charger Budget Sweep Reveals P_3 Saturation



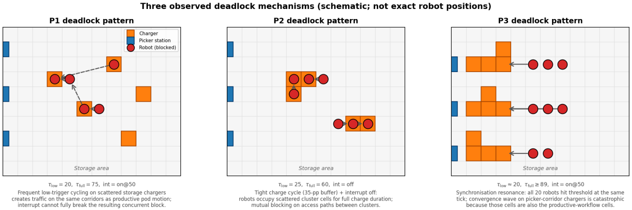


Budget sweep ($T = 20\,000$ s).

- P_1, P_2 essentially flat across budgets — adding chargers does not help.
- P_3 climbs from 23 % $\rightarrow$ 60 % at $|C| = 50$.
- Marginal return drops sharply beyond $|C| = 35$.
- Saturation of the picker-tier heuristic.

Cross-pipeline pattern: placement geometry dominates. The threshold-policy lever exists but is an order of magnitude smaller than the placement lever.

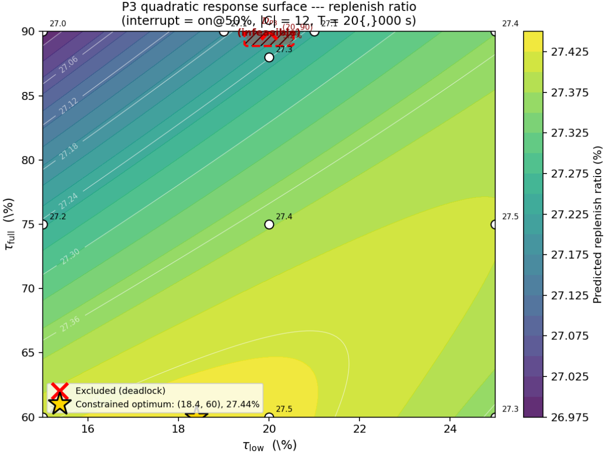
Phase 1 — Three Deadlock Mechanisms Observed



Three deadlock failure modes observed in the run logs:

1. P_2 *contention wave:* aggressive trigger ($\tau_{low} = 25$, no interrupt) sends too many robots to charge at once.
2. P_3 *resonance:* interrupt at 50 % releases robots in synchronised waves that re-collide at picker stations.
3. P_1 *corridor block:* chargers placed in transit aisles create chokepoints when fleet is depleted.

These three patterns motivated the Phase 2 constrained-RSM treatment.

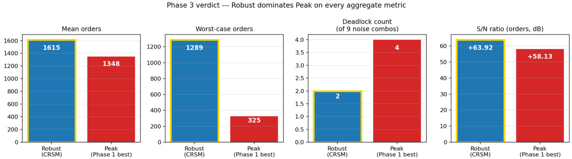


Fitted quadratic surface (P_3 replenish ratio).

- Thin deadlock zone discovered at $\tau_{\text{low}} \approx 20\,\%$, $\tau_{\text{full}} \geq 89\,\%$.
- Treated as infeasibility constraint — not missing data.
- **Constrained optimum:** $\tau_{\text{low}} = 18\,\%$, $\tau_{\text{full}} = 60\,\%$, interrupt on @ $50\,\%$.

Phase 2b saturation fit: Michaelis–Menten on $|C|/N_r$ shows P_3's knee at $|C|/N_r \approx 0.5$ — justifies $|C| = 12$ as the CapEx anchor.

Phase 3 — Robust Beats Peak on Mean *and* Robustness

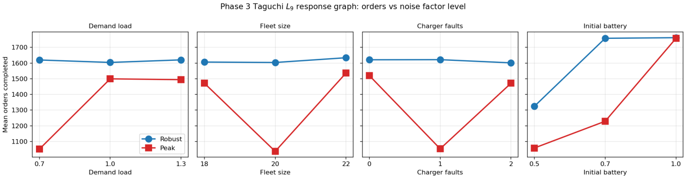


Metric	Peak	Robust
Mean orders (L_9)	1,348.1	**1,614.7**
Deadlocked cells	4/9	**2/9**
Orders S/N (dB)	—	**+5.79**
η_E (kJ/order)	48.8	**44.4**

Candidates compared:

- *Peak:* Phase-1 best corner (interrupt off).
- *Robust:* Phase-2 constrained-RSM optimum.

Robust dominates across *every* aggregate metric — mean, robustness, deadlock, and energy cost per order.



Taguchi response graph: order count vs level for each noise factor.

- *Initial battery:* largest swing across levels.
- Demand swing $\pm 30\,\%$: absorbed.
- One to two charger faults: absorbed.
- Fleet attrition ± 2: absorbed.

Counter-intuitive insight: the controllable lever is *not* demand or fleet size — it is start-of-shift battery state. The procedural fix is a shift-handover protocol, not a demand-forecasting investment.

Decision-Grade Output — CapEx, OpEx, Procedural

Tier	Recommendation	Justification
CapEx	P_3 placement; $\|C\| = 12$ chargers in picker corridor	$30\times$ replenish margin over $P_1/P_2/P_4$; saturation knee below $\|C\| = N_r$
OpEx	$\tau_{\text{low}} = 18\,\%$, $\tau_{\text{full}} = 60\,\%$, interrupt on @ $50\,\%$	Phase-2 constrained-RSM optimum; confirmed as robust optimum in Phase 3
Procedural	Ensure $\geq 70\,\%$ start-of-shift battery	Phase-3 dominant noise factor; 50 % start state risks deadlock

Each tier is traceable to a specific experimental phase, and each decision is defended by the corresponding statistical machinery: response surfaces (Phase 2), signal-to-noise ratios (Phase 3), deadlock-manifold characterisation (Phase 1–2).

Financial Projection — Solusi Memiliki Fundamental Finansial yang Kokoh

Estimasi Parameter

Operating Profile
2 shifts × 4,800 hr/yr

Electricity (Taipower)
NT$3.50/kWh

Margin per Pick
NT$15/order (grocery)

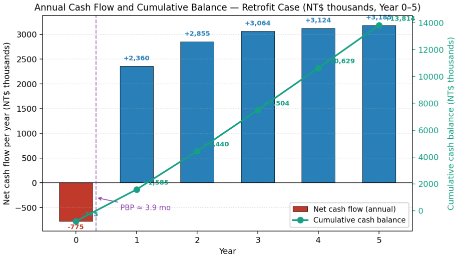
Charger Cost
NT$155k + NT$46.5k install

Relocation
NT$37.2k per unit

Tax / WACC
20% / 10%

Battery Life ↑
+30% at τ_{full}=60%

Currency
NT$31 = USD$1



Net Cash Flow (bars) & Cumulative Balance (line), Retrofit case

Proyeksi Finansial

3.9 mo
Payback Period

304%
5-yr IRR

NT$10.1 M
5-yr NPV @ 10%

NT$2.9–4.0 M
Annual revenue gain

204%
Year-1 ROI

Verdict

Tren positif yang signifikan menandakan solusi **layak dijalankan.**

Financial Projection — CapEx & OpEx Breakdown

CapEx (Year 0 — One-Time Investment)

Item	Qty	NT$ (thousands)
Industrial charger unit	2	310.0
Installation (anchor + wiring)	2	93.0
Relocation of existing chargers	10	372.0
Total CapEx		**775.0**

Depreciation (5-yr straight-line)

$$D = \frac{\text{NT\$775k}}{5} = \text{NT\$155k/yr}$$

Non-cash; reduces taxable income ⇒ tax shield $D \cdot t = \text{NT\$31k/yr}$.

OpEx Delta (Year 1, before ramp & inflation)

Item	Type	NT$/yr (thousa
Revenue gain (244,320 picks × NT$15)	+	3,6
Energy saving (2,630 kWh × NT$3.50)	+	
Battery longevity *(excluded — not modelled)*	—	
Maintenance (7% × new CapEx NT$403k)	−	
Pre-tax cash benefit (full-ramp)		**3,6**

Year-on-Year Dynamics

- Ramp: 80% / 95% / 100% / 100% / 100%
- Revenue inflation: +2%/yr (CPI pass-through)
- Energy inflation: +3%/yr (Taipower trend)
- Maintenance escalation: +5%/yr

Financial Projection — 5-Year Cash Flow Projection

Pre-Tax Income (NT$ thousands)

Year	1	2	3	4	5
Revenue gain	2,931.8	3,551.0	3,812.8	3,889.1	3,966.9
Energy save	7.4	9.0	9.8	10.1	10.4
Maintenance	−28.2	−29.6	−31.1	−32.6	−34.3
Pre-tax	**2,911.0**	**3,530.4**	**3,791.5**	**3,866.6**	**3,943.0**

Battery longevity excluded — degradation not modelled (Limitations).

Net Income & After-Tax Cash Flow (NT$ thousands)

Year	1	2	3	4	5
Pre-tax	2,911.0	3,530.4	3,791.5	3,866.6	3,943.0
Depreciation	−155.0	−155.0	−155.0	−155.0	−155.0
Taxable income	2,756.0	3,375.4	3,636.5	3,711.6	3,788.0
Tax (20%)	−551.2	−675.1	−727.3	−742.3	−757.6
ATCF	**2,359.8**	**2,855.3**	**3,064.2**	**3,124.3**	**3,185.4**
Cumul.	**1,584.8**	**4,440.1**	**7,504.4**	**10,628.7**	**13,814.1**

Perhitungan Proyeksi

Revenue$_y$ $= \Delta N \cdot m \cdot \rho_y \cdot (1+\iota_R)^{y-1}$

ΔN = 244,320 picks/yr at full ramp

Energy save$_y$ $= \Delta E \cdot p_e \cdot \rho_y \cdot (1+\iota_E)^{y-1}$

Maintenance$_y$ $= -0.07 \cdot \text{CapEx}_{\text{new}} \cdot (1+\iota_M)^{y-1}$

Pre-tax$_y$ = sum of above

Tax$_y$ $= -t \cdot (\text{Pre-tax}_y - D)$

ATCF$_y$ $= \text{Pre-tax}_y + \text{Tax}_y$

PBP = interpolated zero-crossing of cumulative ATCF

NPV$_5$ $= -\text{CapEx} + \sum_{y=1}^{5} \frac{\text{ATCF}_y}{(1+r)^y}$

IRR: solve $\sum \frac{\text{ATCF}_y}{(1+\text{IRR})^y} = \text{CapEx}$

Battery saving deliberately excluded.

Headline

PBP = 3.9 months
NPV$_5$ = NT\$10,144k
IRR = 304%
Year-1 ROI = 204%

The RL Extension — A Documented Negative Result

Why we tried RL. The literature (Bischoff et al. 2025, Singh et al. 2024) reports that learned policies can refine threshold-based heuristics by a few percent. We wanted to test whether the same held on our environment.

Setup. MaskablePPO (Proximal Policy Optimization with action masking; sb3-contrib). Per-AMR-per-event decisions; 9 discrete charge targets; 15-dim observation. Four configurations exercised on P_1.

Outcome. All four runs collapsed within 4–8 PPO iterations: policy entropy $\to 0$, action mass concentrated on action 0 ("no charge"), mean reward decayed monotonically as the all-action-0 policy drove robots into depletion.

Three structural root causes:

1. *Action-mask asymmetry.* Action 0 is always legal; charge actions require $t > b$.
2. *Reward myopia.* Per-decision reward Δorders pays off immediately; battery-depletion penalty fires ~ 1000 ticks later — beyond PPO's $\gamma = 0.99$ effective horizon (~ 100 steps).
3. *Decision-granularity asymmetry.* Action 0 returns the agent to the next event quickly; charge actions silence the focal robot for many ticks $\Rightarrow$ sparser reward signal.

Implication: the collapse *strengthens* the methodological choice. Fixing it requires re-engineering the env in ways that decouple RL from the heuristic baseline it would refine — partly defeating the original motivation.

Limitations and Future Work

Limitations of the present study:

- Single-seed deterministic protocol (no pure-error variance estimate from replicates).
- Policy axis restricted to the three-knob threshold class ($\tau_{\text{low}}, \tau_{\text{full}}$, interrupt).
- Deadlock manifolds characterised within the immediate neighbourhood of the design centre; wider sweeps may reveal additional infeasible regions.
- No regenerative braking, no charging curve, perfectly flat warehouse floor.

Future work:

- Multi-seed replication of Phase 2 and Phase 3 for pure-error variance estimation.
- Hybrid placements combining P_3's dwell-alignment with P_1/P_2's spatial spread.
- Capacity-vs-count comparison: raising E_{cap} instead of $|C|$.
- Re-engineered RL environment (dense battery-cost shaping, null-action-free space, fleet-level decisions) to test whether a learned policy can break the depletion ceiling.

Thank You.

Questions & Discussion

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